

Example 6 : The electric potential V at a point $P(x, y, z)$ in space is given by $V = 4x^2$ volt. Electric field at point (1m, 0, 2m) in V/m is

(a) 8 along -ve x-axis

(b) 8 along +ve x-axis

(c) 16 along -ve x-axis

(d) 16 along +ve x-axis

Solution:

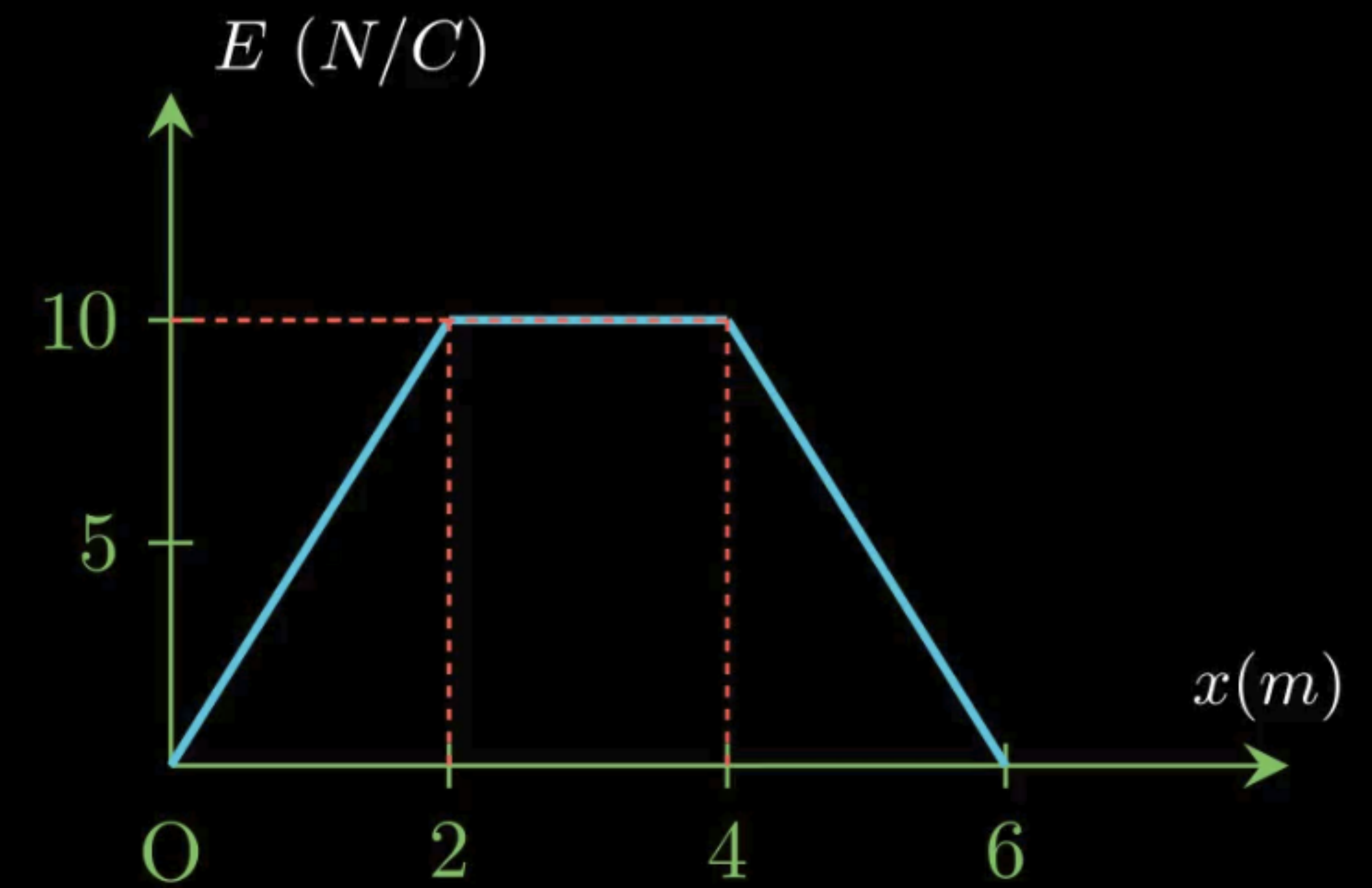
Example 7 :Figure shows the variation of electric field intensity E versus distance x . What is the potential difference between the points at $x = 2$ m and $x = 6$ m from O?

(a) 30 V

(b) 60 V

(c) 40 V

(d) 80 V



Example 8 : An infinite plane sheet of charge density 10^{-8} Cm^{-2} is held in air. In this situation how far apart are two equipotential surfaces, whose potential difference is 5 V?

(a) 2.25 mm

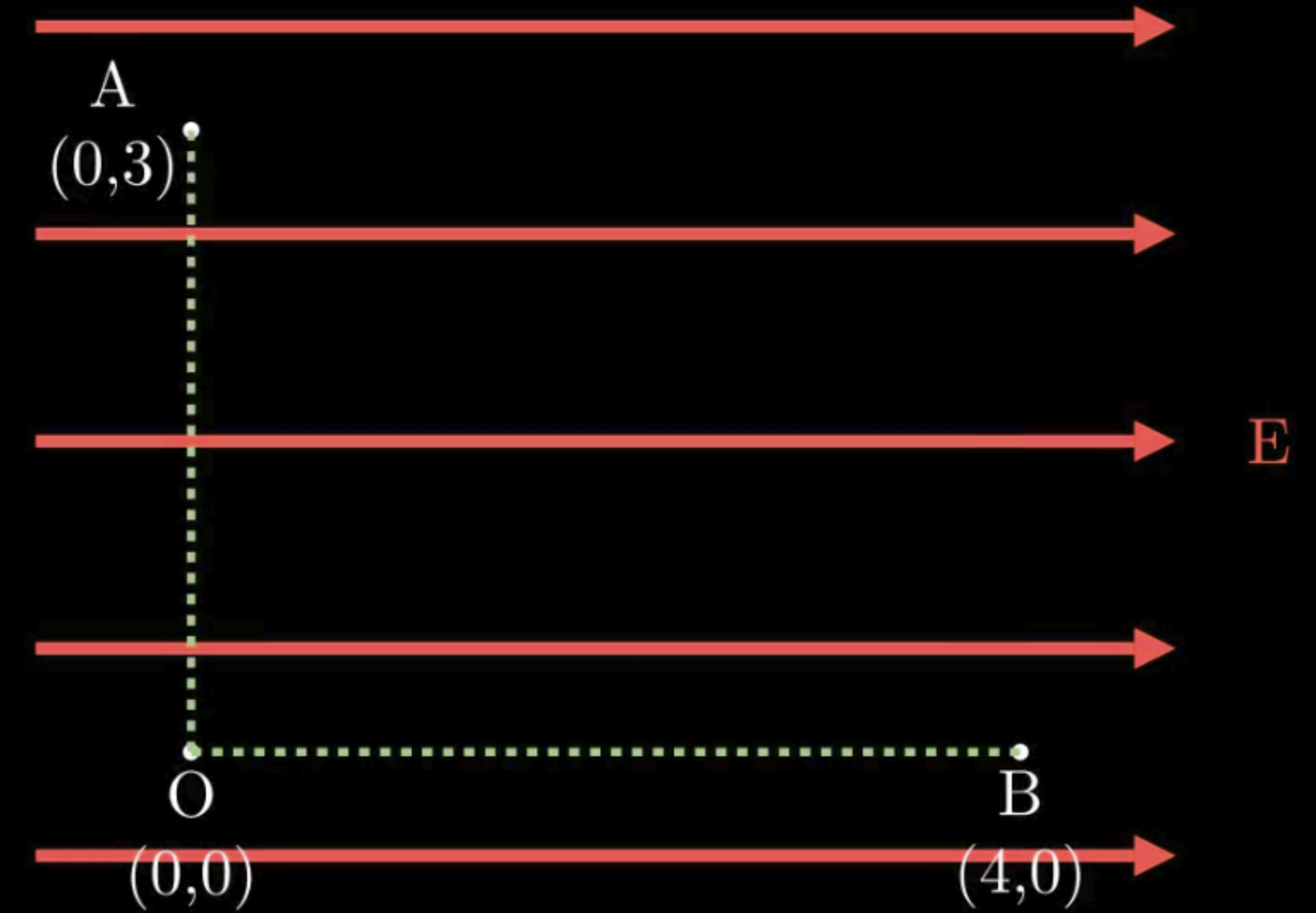
(b) 3.52 mm

(c) 6 mm

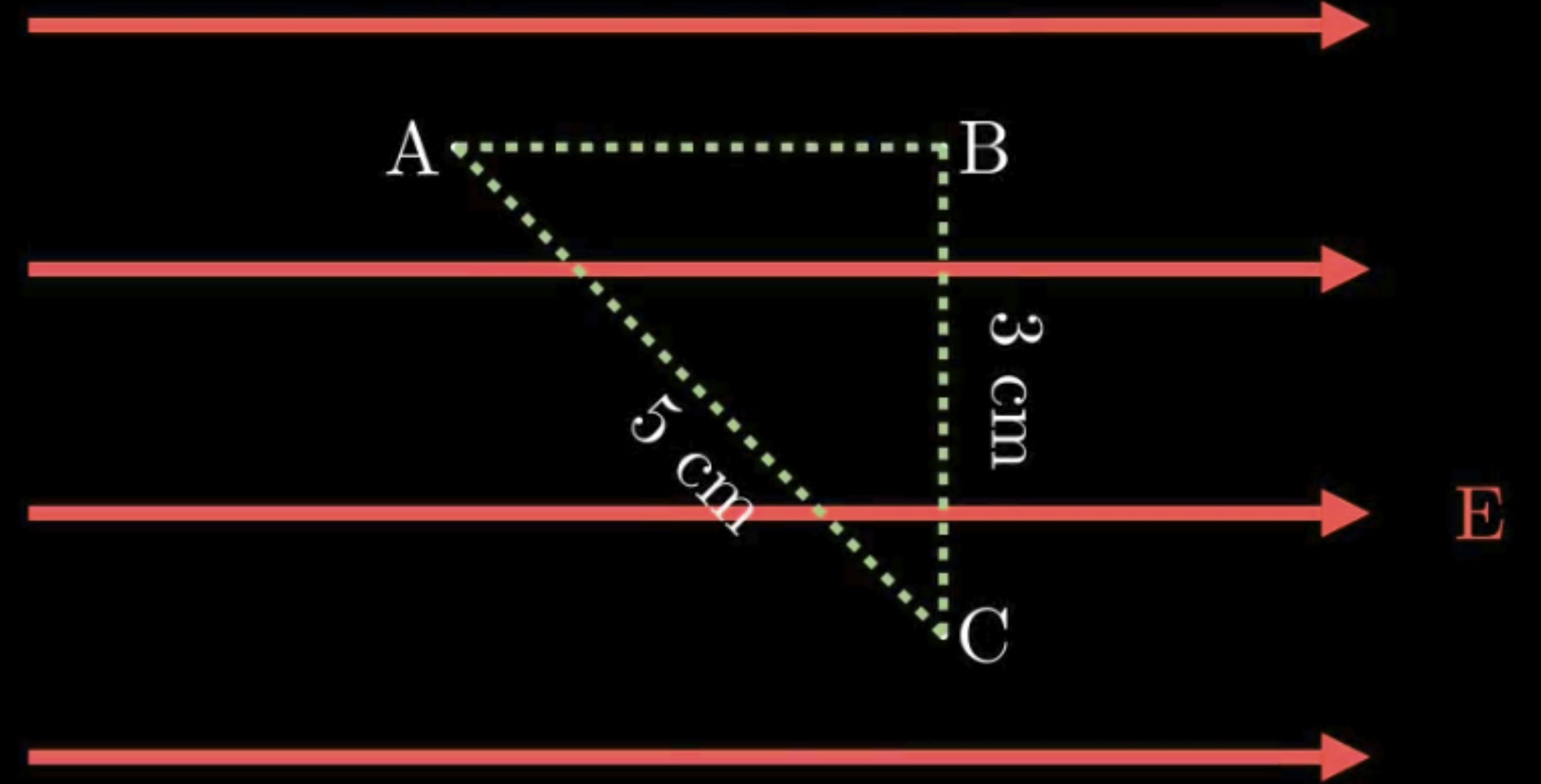
(d) 8.85 mm

Example 9 : A uniform electric field E of 500 N/C is directed along $+x$ -axis. O, B and A are three points in the field having x- and y-coordinates (in cm) (0, 0), (4, 0) and (0, 3) respectively. Calculate the potential difference between the points (i) O and A, and (ii) O and B. [CBSE 23C]

Solution:



Example 10 : Three points A, B and C lie in a uniform electric field (E) of $5 \times 10^3 \text{ NC}^{-1}$ as shown in the figure. Find the potential difference between A and C. [CBSE F 09]



Example 11 : If the potential in the region of space around the point $(-1 \text{ m}, 2 \text{ m}, 3 \text{ m})$ is given by $V = (10x^2 + 5y^2 - 3z^2)$ volt, calculate the three components of electric field at this point.

Solution:

- 31.** (a) (i) The electric field in a region is given by $\vec{E} = 40x \hat{i}$ N/C. Find the amount of work done in taking a unit positive charge from a point (0, 3m) to the point (5m, 0).

(a)

$$\begin{aligned} \text{(i)} \quad V &= -\int \vec{E} \cdot \vec{dr} \\ &= -\int 40x \, dx \\ &= -20x^2 \end{aligned}$$

Potential at A (0, 3m), $V_A = 0$

Potential at B (5m, 0), $V_B = -500 \, \text{V}$

Work done in taking a unit positive charge from a point (0, 3m) to the point (5m, 0)

$$\begin{aligned} W &= q(V_B - V_A) \\ &= 1(-500 - 0) \end{aligned}$$

$$W = -500 \, \text{J}$$

1/2

1/2

1/2

1/2

- (ii) Write two characteristics of equipotential surfaces. A uniform electric field of 50 NC^{-1} is set up in a region along $+x$ axis. If the potential at the origin $(0, 0)$ is 220 V , find the potential at a point $(4\text{m}, 3\text{m})$.

(ii) Characteristics of equipotential surfaces: -
(Any two)

- Potential at all points on the surface is same.
- Equipotential surface is normal to the direction of the electric field.
- The work done in moving a charge on an equipotential surface is zero.

$$V_0 - V = E d = 50 \times 4$$

$$V_0 - V = 200 \text{ V}$$

$$V = 220 \text{ V} - 200 \text{ V}$$

$$V = 20 \text{ V}$$

$$\frac{1}{2} + \frac{1}{2}$$

$$\frac{1}{2}$$

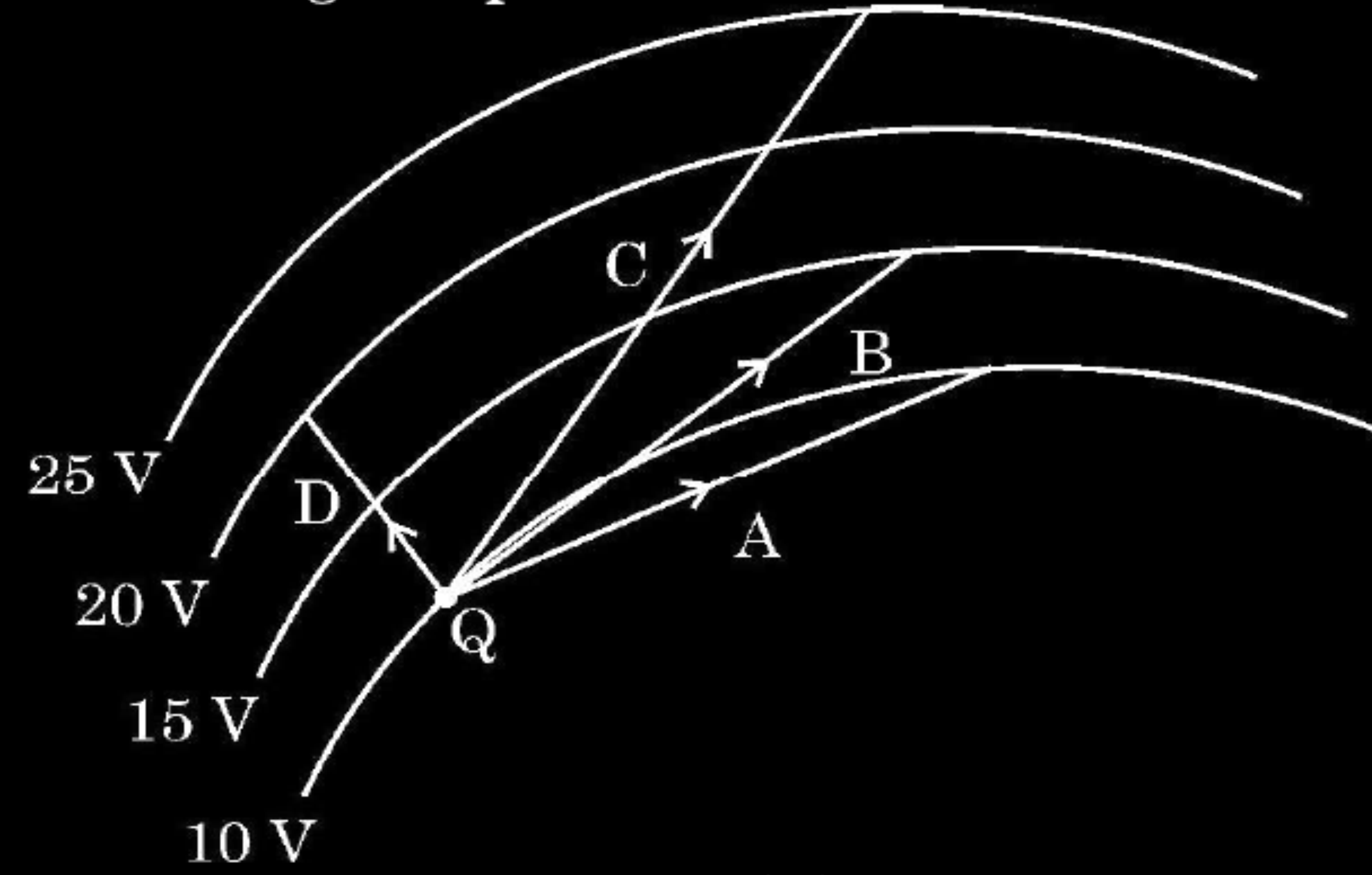
$$\frac{1}{2}$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$

1. In the figure curved lines represent equipotential surfaces. A charge Q is moved along different paths A, B, C and D. The work done on the charge will be maximum along the path

1



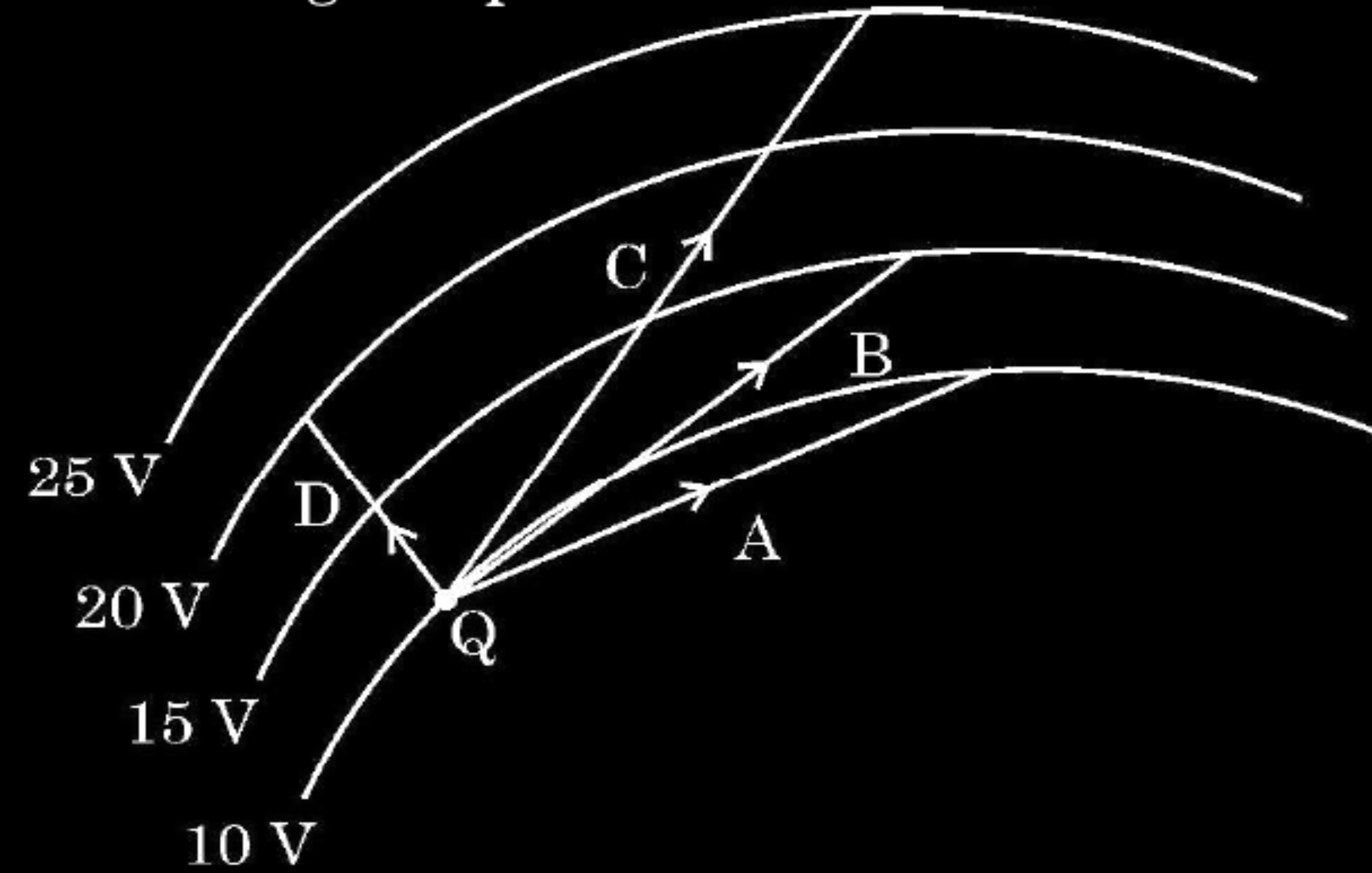
(A) A
(C) C

(B) B
(D) D

CBSE 25

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(D) D

CBSE 25

(C) C

29. The electric potential at a point in an electric field is the work done in bringing a unit positive charge from infinity to this point. If the potential difference between any two points at a surface is zero, the surface is called an equipotential surface. The shape of an equipotential surface can give direction of electric field at a point on it. It may be a closed surface or an open surface, depending on the charges creating the electric field.

(i) The shape of equipotential surface due to an isolated point charge is :

- (A) ellipsoidal, with charge at its one foci
- (B) plane surface not passing through point charge
- (C) spherical with charge at its centre
- (D) cylindrical with charge at its axis

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CBSE 24 C

(C) Spherical with charge at its centre

(ii) The equipotential surfaces in a uniform electric field acting along + Z direction are :

1

- (A) planes parallel to the XY plane
- (B) concentric spherical surfaces
- (C) planes parallel to the YZ plane
- (D) planes parallel to the XZ plane

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(A) Planes parallel to the XY plane

(iii) Which of the following statements is *not* true for equipotential surfaces ?

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- (A) Potentials at two equipotential surfaces are different.
- (B) No work is done in moving a charge on an equipotential surface.
- (C) Equipotential surfaces always enclose some volume.
- (D) Two equipotential surfaces do not intersect each other.

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(C) Equipotential surfaces always enclose some volume

- (iv) (a) The angle between the electric field at a point and outward normal at that point on an equipotential surface is : *1*
- (A) 90° (B) 45°
(C) 30° (D) 0°

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(A) 90°

(B) 45°

(C) 30°

(D) 0°

(D) 0^0

(iv) (b) A and B are two equipotential surfaces around a point charge Q. Surface A is closer to the charge than surface B. If a test charge is released between them, it will : 1

- (A) remain stationary
- (B) move from A to B
- (C) move from B to A
- (D) move around Q in a circular path between the equipotential surfaces A and B

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- (D) move around Q in a circular path between the equipotential surfaces A and B

(B) move from A to B

1. Consider a group of charges $q_1, q_2, q_3 \dots$ such that $\Sigma q \neq 0$. Then equipotentials at a large distance, due to this group are approximately :

(A) Plane

(B) Spherical surface

(C) Paraboloidal surface

(D) Ellipsoidal surface

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CBSE 24

(B) Spherical surface

- 24.** (a) A uniform electric field E of 500 N/C is directed along $+x$ axis. O, B and A are three points in the field having x and y coordinates (in cm) $(0, 0)$, $(4, 0)$ and $(0, 3)$ respectively. Calculate the potential difference between the points (i) O and A, and (ii) O and B.

2

CBSE 23 C

- 24.** (a) A uniform electric field E of 500 N/C is directed along $+x$ axis. O, B and A are three points in the field having x and y coordinates (in cm) (0, 0), (4, 0) and (0, 3) respectively. Calculate the potential difference between the points (i) O and A, and (ii) O and B.

2

CBSE 23 C

(a) (i) $V_{OA} = E (x_2 - x_1)$ $V_{OA} = 500 \times 0 = 0$ volt	$\frac{1}{2}$ $\frac{1}{2}$
(ii) $V_{OB} = -E (x_2 - x_1)$ $V_{OB} = -500 \times (4 \times 10^{-2})$ $= -20$ V	$\frac{1}{2}$ $\frac{1}{2}$

37. (a) Write two important characteristics of equipotential surfaces.
- (b) The magnitude of electric field (in NC^{-1}) in a region varies with the distance r (in m) as

$$E = 10r + 5$$

CBSE 20

By how much does the electric potential increase in moving from point at $r = 1$ m to a point at $r = 10$ m.

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CBSE 20

By how much does the electric potential increase in moving from point at $r = 1$ m to a point at $r = 10$ m.

- (a) For equipotential surfaces
- (i) Potential has the same value at all points on the surface.
 - (ii) Electric field is normal to the equipotential surface at all points
 - (iii) Work done in moving any charge between any two points on the equipotential surface is zero (any two)

1+1

(b)

$$E = 10r + 5$$

$$dV = -E \cdot dr$$

$$\int dV = - \int_1^{10} \vec{E} \cdot d\vec{r}$$

$$= - \int_1^{10} (10r + 5) dr$$

$$V = - \left[\int_1^{10} 10r dr + \int_1^{10} 5 dr \right]$$

$$V = 10 \left[\frac{r^2}{2} \right]_1^{10} + 5(r)_1^{10}$$

$$V = -5[100 - 1] + 5[10 - 1]$$

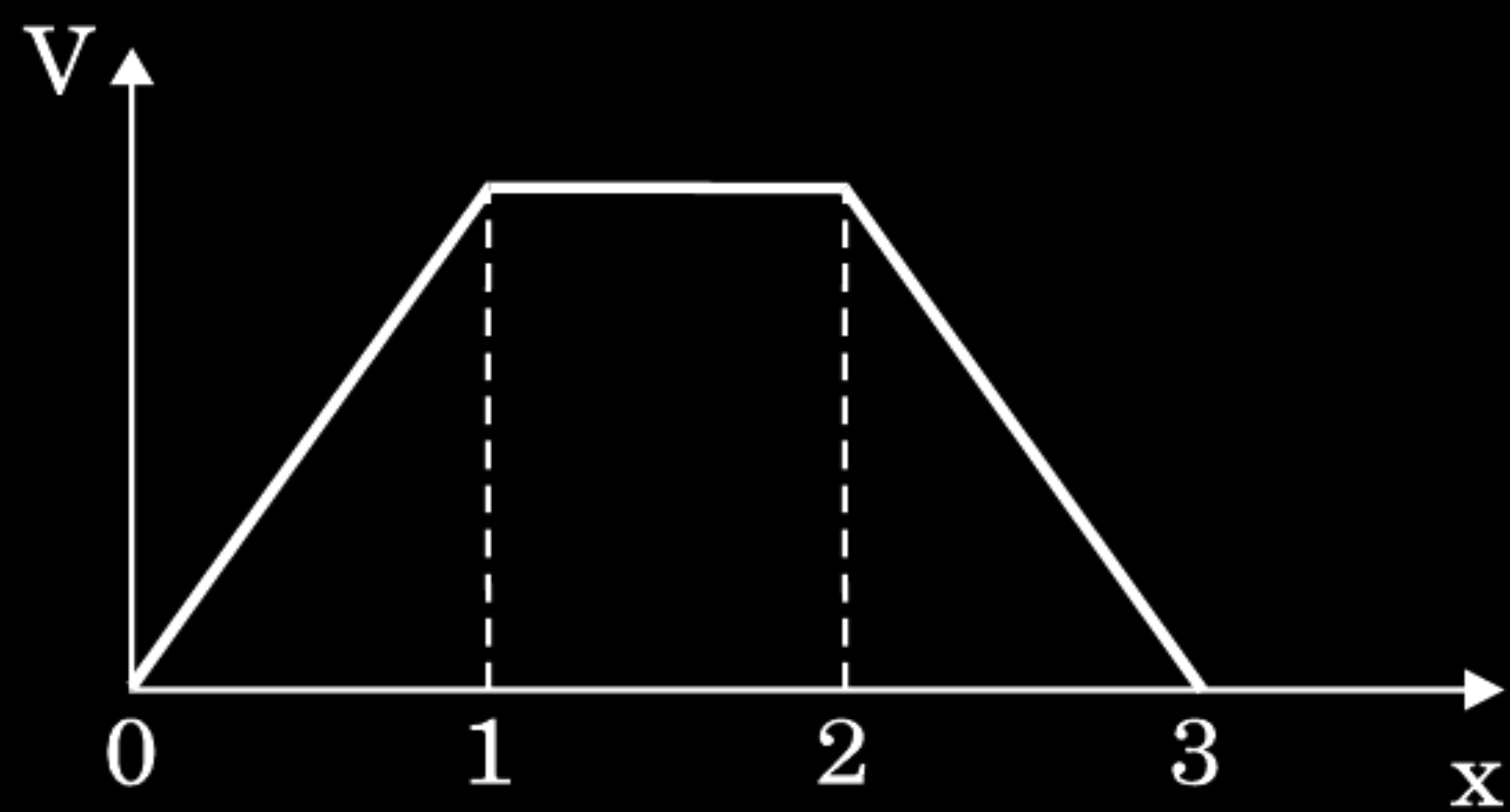
$$V = -5 \times 99 + 5 \times 9 = -540V$$

$\frac{1}{2}$

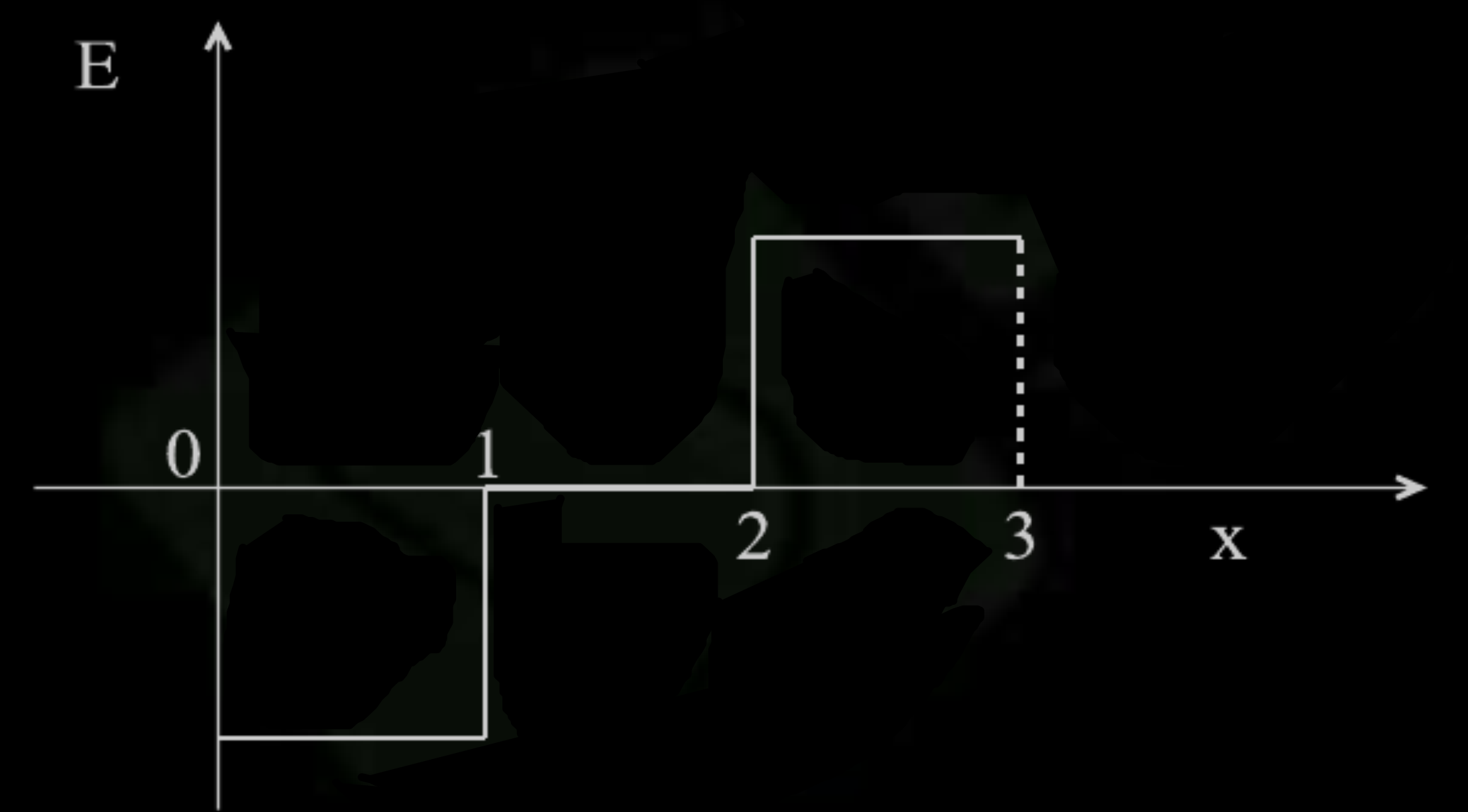
$\frac{1}{2}$

$\frac{1}{2}$

- (c) The electric potential as a function of distance 'x' is shown in the figure. Draw a graph of the electric field E as a function of x .



iii)



Draw equipotential surfaces for an electric dipole.

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CBSE 19



(Even if a student mentions or draws equatorial plane, award 1 mark.)

1

Define an equipotential surface. Draw equipotential surfaces :

- (i) in the case of a single point charge and
- (ii) in a constant electric field in Z-direction.

Why the equipotential surfaces about a single charge are not equidistant ?

- (iii) Can electric field exist tangential to an equipotential surface ? Give reason.

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Surface with a constant value of potential at all points on the surface.

ii. 

$V \propto \frac{1}{r}$

iii.No

If the field lines are tangential, work will be done in moving a charge on the surface which goes against the definition of equipotential surface.

1/2

1/2

1/2

1/2

